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003602

August/September 2022

B.Tech. (CE/CSE) VI SEMESTER

Computer Networks (PCC-CS-602)

Time : 3 Hours] [Max. Marks : 75

Instructions :

1. *It is compulsory to answer all the questions (1.5 marks each) of Part-A in short.*
2. *Answer any four questions from Part-B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART-A

1. (a) Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference? (1.5)

(b) What is the significance of the twisting in twisted-pair cable? (1.5)

(c) A code scheme has a Hamming distance dmin == 4. What is the error detection and correction capability of this scheme? (1.5)

Question paper

(d) How is the preamble field different from the SFD field? (1.5)

(e) What are the shortcomings of class-full addressing? (1.5)

(f) Which is better DHCP or BOOTP and why? (1.5)

(g) How TCP protocol handles congestion? (1.5)

(h) Name the policies that can prevent congestion. (1.5)

(i) How iterative resolution is different from recursive resolution? (1.5)

(j) How HTTP is same as well as different from SMTP and FTP? (1.5)

PART-B

List out the protocols, data units and communicating devices operating on each layer of OSI Model. (7)

Define spread spectrum and its goal. List any two spread spectrum techniques. (5)

Why is there no need for CSMAlCD on a full-duplex Ethernet LAN? (3)

Find the final code after encoding data bits 1101 into 7-bit even parity Hamming Code? (6)

Why the size of the sender and receiver windows must be at most onehalf of 2 m in selective Repeat ARQ? Explain with an appropriate example. (4)

Compare CSMA/CD with CSMA/CA. (5)

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Question paper

OR

(d) One hundred stations on a pure ALOHA network share a 1-Mbps channel. If frames are 1000 bits long, find the throughput if each station is sending 10 frames per second. (5)

4. (a) An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows: (i) The first group has 64 customers; each needs 256 addresses. (ii) The second group has 128 customers; each needs 128 addresses. (iii) The third group has 128 customers; each needs 64 addresses. Design the sub-blocks and find out how many addresses are still available after these allocations. (10)

(b) List RIP shortcomings and their corresponding fixes. (5)

5. (a) Why transport layer is called an end-to-end layer? Draw a TCP State transition diagram. (5)

(b) What is the value of the receiver window (rwnd) for host A if the receiver, host B, has a buffer size of 5000 bytes and 1000 bytes of received and unprocessed data? Also explain any two techniques to shape traffic? (10)

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6. (a) How IMAP is better than POP3 Protocol? (5)

(b) Explain Request, Response and Entity Headers used in HTTP. (10)

7. Write short note on following :

(i) ALOHA.

(ii) SCTP.

(iii) Firewalls. (15)

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016405

May 2024

B.Tech. (CEDS) (Fourth Semester)

Computer Networks (PCC-CS-602)

Time : 3 Hours] [*Maximum Marks : 75*

Note : It is compulsory to answer all the questions (1.5 marks each) of Part A in short. Answer any *four* questions from Part B in detail. Different sub-parts of a question are to be attempted adjacent to each other.

Part A

1. (a) What is piggybacking ? 1.5

   (b) What are the number of cables required in mesh topology ? 1.5

   (c) The subnet mask for a particular network is 255.255.31.0. Which of the following pairs of IP addresses could belong to this network ? 1.5

   (i) 172.57.88.62 and 172.56.87.233

   (ii) 10.35.28.2 9.4

(iii) 191.203.31.87 and 191.234.31.88

(iv) 128.8.129.43 and 128.8.161.55

(d) DSSS system spreads the baseband signal by.....the baseband pulses with a pseudo noise sequence. **1.5**

(e) What is the purpose of IEEE 802.11 specifications ? **1.5**

(f) A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time (including the delays in the devices and ignoring the time needed to send a jamming signal) is 25.6 μs, what is the minimum size of thcce frame ? **1.5**

(g) Station A uses 32 byte packets to transmit messages to Station B using a sliding window protocol. The round trip delay between A and B is 80 milliseconds and the bottleneck bandwidth on the path between A and B is 128 kbps. What is the optimal window size that A should use ? **1.5**

(h) What is the purpose of DHCP protocol ? **1.5**

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(i) What is the purpose of SNMP protocol ? 1.5

(j) What is the difference between physical address and logical address ? 1.5

Part B

2. (a) What is the minimum hamming distance of the following strings 010, 011, 101 and 111. 10

(b) Explain the FHSS technique for data transmission. 5

3. (a) Explain the CRC coding and decoding mechanism by taking example of 1001 as dataword in C(7, 4) scheme. 5

(b) Assume in stop and wait ARQ system, the bandwidth of the line is 1 Mbps and 1 bit takes 30 ms to make a round trip. What is the bandwidth delay product ? If the system data frames are 1000 bits in length, what is the utilization percentage of the link. 10

4. (a) Explain why the window size should be < 2 m in Go back n ARQ. where m is the size of the sequence number field in bits. 5

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(b) Draw the flowchart of the CSMA/CD protocol. 5

(c) What is the difference between OSI model and TCP/IP model ? 5

5. (a) What is the difference between UDP and TCP ? 5

(b) A slotted ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces : 10

(i) 1000 frames per second

(ii) 500 frames per second

(iii) 250 frames per second.

6. (a) What is the difference between IPv4 and IPv6 addressing ? 10

(b) Explain the basic concepts of cryptography. 5

7. (a) Explain the token bucket algorithm in detail. 5

(b) Write short notes on the following : 10

(i) DNS

(ii) WWW.

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Sr. No. 011501

December 2023

B.Tech (IT/CSE(AIML)) 5th Semester

Computer Network (PCC-CS-602)

Time: 3 Hours Max. Marks:75

Instructions:
1. *It is compulsory to answer all the questions (1.5 marks each) of Part -A in short.*
2. *Answer any four questions from Part -B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART -A

Q1 (a) Categorize four basic topologies in terms of line configuration. (1.5)

(b) What is the difference between Network layer delivery and transport layer delivery? (1.5)

(c) What is minimum Hamming Distance? Specify minimum Hamming distance for detection of three errors. (1.5)

(d) A network using CSMA/CD has a bandwidth of 10 Mbps. If maximum propagation time is 25.6μs, what is the minimum size of the frame? (1.5)

(e) Why ARP is often cited as Security weakness? (1.5)

(f) A block of address is granted to a small organization. One of the addresses is 205.16.37.39/28. What are the first address, last address and total number of addresses? (1.5)

(g) What is the maximum and minimum size of a TCP Header and why? (1.5)

(h) Name some of the policies that can be used to prevent congestion. (1.5)

(i) What are the functions of two FTP Connections? (1.5)

(j) What are Telnet Options? How they are negotiated? (1.5)

PART -B

Q2 (a) What is TCP/IP Model, explain its layers. Also specify the data units, protocols used and communicating devices used at each layer of TCP/IP Model. (7)

(b) Discuss advantage of Optical Fiber over twisted pair and coaxial cable. (3)

(c) Distinguish between multilevel TDM, multiple slot TDM, and pulse-stuffed TDM (5)

Q3 (a) Explain the working of CSMA/CD protocol. (5)

(b) Compare and contrast Go Back N-ARQ with selective Repeat ARQ. Why the size of sender window must be less than 2^m in Go-Back N-ARQ, where m is the size of the sequence number in bits. (10)

Q4 (a) Compare and Contrast IPV4 and IPv6 packet formats. How can we make transition from IPv4 to IPv6. (5)

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(b) An ISP is granted a block of addresses starting with 150.80.0.0/16. The ISP wants to distribute these blocks to customers as follows. (5)

i. The first group has 256 medium-size businesses; each needs 16 addresses.
ii. The second group has 64 small businesses; each needs 8 addresses.
iii. The third group has 128 households; each needs 4 addresses.

Design the sub-blocks and give the slash notation for each sub-block. Find out how many addresses are still available after these allocations.

(c) List the problems associated with Distance Vector Routing and how they can be resolved? (5)

Q5 (a) Explain the characteristics and frame format of TCP protocol. (7

(b) In a connection, the value of cwnd is 3000 and the value of rwnd is 5000. The host has sent 2000 bytes which has not been acknowledged. How many more bytes can be sent? Also explain the techniques used for traffic shaping. (8

Q6 (a) What is HTTP? Explain its request methods and status codes (5

(b) Explain the protocols used for sending and receiving email (5

(c) Differentiate between Proxy and Packet filter Firewall (5

Q7 Write short note on following (any three) (15

i. Standard Ethernet
ii. Pure Aloha and Slotted Aloha
iii. Connectionless Protocols
iv. Symmetric key cryptography

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May 2025

B. Tech. (Sixth Semester)

Computer Networks (PCC-CS-602)

Time : 3 Hours] [*Maximum Marks : 75*

Note : It is compulsory to answer all the questions (1.5 marks each) of Part A in short. Answer any *four* questions from Part B in detail. Different sub-parts of a question are to be attempted adjacent to each other.

Part A

1. (a) Explain the characteristics of twisted pair and coaxial cable. 1.5

(b) Why is layered architecture of network preferred ? 1.5

(c) Describe HDLC's highest link control protocol used for LAN 1.5

(d) Given 1101011011 data frame and generator polynomial $G(x) = x^4 + x + 1$. Derive the transmitted frame. 1.5

(e) Distinguish between ARP and RARP Protocols. 1.5

(f) What is CIDR addressing ? 1.5

(g) Why is transport layer called true end-to-end layer ? 1.5

(h) What are ephemeral ports ? Why are they so called ? What is their range ? 1.5

(i) Differentiate between Persistent and Non-Persistent connection in HTTP. 1.5

(j) What is DDNS ? 1.5

Part B

2. (a) Explain the four basic topologies used in networks. List advantages and disadvantages of each of them. 10

   (b) What is spread spectrum and explain why spreading the signal over a wider frequency band is beneficial. 5

3. (a) Compare various error and flow control protocols of data link layer. 10

   (b) Give a detailed note on the ALOHA protocols. 5

4. (a) An ISP is granted a block of addresses starting with 150.80.0.0/16. The ISP wants to distribute these blocks to 2600 customers as follows. (i) The first group has 200 medium-size businesses; each needs 128

addresses. (ii) The second group has 400 small businesses; each needs 16 addresses. (iii) The third group has 2000 households; each needs 4 addresses. Design the sub-blocks and give the slash notation for each sub-block. Find out how many addresses are still available after these allocations. 5

(b) Show, how to form the least cost tree using Dijkstra Algorithm with an example. Also Describe the problems and solutions associated with distance vector routing. 10

5. (a) Describe key features : multi-streaming, multi-homing, message-oriented service of SCTP. Also Compare SCTP with TCP and UDP in tabular form. 10

(b) Define congestion and its impact on network performance. Explain congestion detection and avoidance strategies in TCP : 5

6. (a) Explain the structure and functioning of the World Wide Web (WWW). What are its core components and protocols ? 10

(b) Explain HTTP methods and HTTP headers. 5

7. Write short notes on the following : (any *three*) 15

(i) Virtual LAN

(ii) DHCP

(iii) Traffic shaping Algorithms

(iv) DNS

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003602

May 2024

B. Tech. (CE/CE(HINDI)) (Sixth Semester)

Computer Networks (PCC-CS-602)

Time : 3 Hours] [*Maximum Marks : 75*

Note : It is compulsory to answer all the questions (1.5 marks each) of Part A in short. Answer any *four* questions from Part B in detail. Different sub-parts of a question are to be attempted adjacent to each other.

Part A

1. (a) Which layer of the OSI model is responsible for dialogue control ? 1.5

(b) What are the advantages and disadvantages of mesh topology ? 1.5

(c) What is Piggybacking ? 1.5

(d) Explain CRC with example. 1.5

(e) What is the use of subnet mask ? 1.5

(f) Identify the class of IP address : 1.5

192.168.2.3

P.T.O.

(g) What are the functions of the transport layer ? 1.5

(h) What parameters affect the QOS ? 1.5

(i) Explain FTP. 1.5

(j) Describe post office protocol. 1.5

Part B

2. (a) Write a note on wireless LAN and explain its frame format. 10

(b) Describe the concept of the Spread spectrum. 5

3. (a) Calculate the hamming distance between code words $C_1 = 10110101$ and $C_2 = 01111010$ 10

(b) Explain CSMA/CD protocol. 5

4. Explain BOOTP and DHCP in detail. 15

5. (a) Compare TCP and UDP with their frame format. 10

(b) Explain the Leaky Bucket protocol. 5

6. A slotted Aloha network transmits 400 bit frames on a shared channel of 400 Kbps. What is life throughput of the system if there are 5 nodes and one system produces 2000 frames/sec. 15

7. Explain the following terms : 15

(a) TELNET

(b) SNMP

(c) Cryptography.

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May 2023

B. Tech. (CE) – VIth Semester

Computer Networks (PCC-CS-602)

Time : 3 Hours]

[Max. Marks : 75

Instructions :

1. *It is compulsory to answer all the questions (1.5 marks each) of Part -A in short.*
2. *Answer any four questions from Part -B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART–A

1. (a) What are the responsibilities of data link layer? (1.5)

   (b) Why wireless LAN's cannot implement CSMA/CD? (1.5)

   (c) Suppose a computer sends a frame to another computer on a bus topology LAN. The physical destination address of the frame is corrupted during the transmission. What happens to the frame? How can the sender be informed about the situation? (1.5)

   (d) How does VLAN reduce network traffic? (1.5)

   (e) What is vulnerable time? (1.5)

(f) What is an allycast address? (1.5)

(g) Differentiate between BOOTP and DHCP? (1.5)

(h) What is the minimum and maximum size of a TCP Header? (1.5)

(i) How HTTP is same as well as different from SMTP? (1.5)

(j) How are options negotiated in TELNET? (1.5)

PART–B

2. (a) Which, layers provide following services :

(i) Route determination.

(ii) Flow Control.

(iii) Interface to transmission media.

(iv) Login-Logout procedures.

(v) Format and Code conversion services. (5)

(b) What is spread spectrum? What are its design goals? (5)

(c) Explain frame format of Ethernet. (5)

3. (a) Given the dataword 1010011110 and the divisor 10111. Show the generation of the codeword at the sender site (using binary division). (5)

(b) Differentiate between Aloha and Slotted Aloha? One hundred stations on a pure ALOHA network share a 1-Mbps channel. If frames are 1000 bits long, find the throughput if each station is sending 10 frames per second. (5)

(c) Explain Go-Back N ARQ flow control protocol for noisy channel. (5)

4. (a) What restrictions are imposed by Internet Authorities on classless address blocks? An ISP is granted a block of addresses starting with 120.60.4.0/22. The ISP wants to distribute these blocks to 100 organizations with each organization receiving just eight addresses. Design the subblocks and give the slash notation for each subblock. Find out how many addresses are still available after these allocations. (5)

(b) What is Link state routing? Explain with the help of example what are the four types of links defined in OSPF? (10)

5. (a) What is sliding window? How does it helps in congestion control? What is the size of the window for a host A if value of rwnd is 3000 bytes and value of cwnd is 3500 bytes? (5)

(b) Prepare a state chart diagram of TCP. Also Explain how TCP handles Congestio Control ? (10)

6. (a) Explain Message Transfer and Message Access protocols in detail. (7)

(b) What is cryptography? Explain various cryptographic algorithms. (8)

7. Write short notes on following (any *three*) :

(i) Transmission Media.

(ii) CSMA/CD.

(iii) Traffic Shaping Techniques

(iv) DNS. (15)

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